

Edge AI Robot for Project DRISHTI

Autonomous Quarry Measurement Robot with Depth Vision

1. Overview

This document specifies how the Edge AI Robot platform is adapted for **Project DRISHTI** — transforming the mobile robot into an autonomous granite block measurement unit that navigates quarry sites, detects blocks, measures dimensions using hardware stereo depth, and calculates seigniorage fees automatically.

Core Value: DRISHTI currently relies on manual mobile phone photography with monocular depth estimation (Depth Anything V2). The robot provides **hardware stereo depth** via OAK-D Pro, giving direct metric measurements accurate to $\pm 2\text{cm}$ — far superior to monocular AI estimation. It also eliminates human effort by autonomously surveying quarry blocks.

2. DRISHTI Requirements Mapped to Robot Hardware

DRISHTI Requirement	Standard Approach (Mobile App)	Robot Adaptation
Granite block detection	YOLOv11 on cloud from phone photos	YOLOv11 on Jetson (edge, real-time, no upload needed)
Depth estimation	Depth Anything V2 (monocular, AI-estimated)	OAK-D Pro stereo depth (hardware, millimeter-accurate)
Reference calibration marker	Government-issued calibration board	Robot-mounted laser reference + stereo baseline (self-calibrating)
Dimension estimation	Geometric analysis from single phone photo	Multi-angle stereo reconstruction (robot drives around block)
GPS tagging	Phone GPS	RTK GPS module on robot (centimeter accuracy)
Image quality validation	Phone flash guidance	Controlled lighting (robot-mounted LED panels)
Volume calculation	$L \times W \times H$ from estimated dimensions	Point cloud reconstruction → accurate convex hull volume
Classification (Above/Below Gangsaw)	From computed volume	Same — automated from measured volume
Seigniorage fee computation	Rules engine on cloud	Same rules engine, runs on Jetson locally

3. Adapted Hardware Configuration

3.1 Compute — NVIDIA Jetson Orin Nano Super (₹21,000)

Running the DRISHTI measurement pipeline: - **YOLOv11** block detection + instance segmentation (TensorRT, 60+ FPS) - **Stereo depth processing** from OAK-D (real-time depth maps) - **Point cloud generation** for 3D volume estimation - **OpenCV geometric analysis** for dimension extraction - **Classification** + **fee computation** engine (on-device)

3.2 Camera — Luxonis OAK-D Pro (₹16,800)

Why stereo depth beats monocular estimation for DRISHTI:

Factor	Monocular (Depth Anything V2)	Stereo (OAK-D Pro)
Accuracy	±10-20cm typical	±1-3cm at 2m range
Calibration needed	Yes (reference marker required)	Self-calibrated (known stereo baseline = 7.5cm)
Lighting dependency	High (affected by shadows, wet surfaces)	Low (IR structured light available)
Metric output	Relative depth → needs scaling	Absolute depth in millimeters directly
Multi-surface handling	Struggles with irregular shapes	Point cloud captures full 3D geometry

OAK-D Pro specifications for quarry use: - Stereo depth range: 0.2m to 35m (covers all granite block sizes) - IR dot projector for structured light in shadowed areas - 4K RGB for visual documentation - Operates in 0°C to 50°C (suitable for AP quarry temperatures)

3.3 Additional Sensor — Laser Distance Sensor (₹2,500)

- TFMini-S or VL53L5X Time-of-Flight sensor
- Cross-validates stereo depth measurements
- Provides ground-truth calibration point for each scan
- Range: 0.1m to 12m, accuracy ±1cm

3.4 Mobility Platform — Tracked Chassis (₹10,000)

Changed from 4WD wheels to tracked/tank chassis for quarry terrain: - Handles loose gravel, dust, uneven rock surfaces - Climbs over small debris (up to 5cm obstacles) - Better traction on dusty/wet quarry floor - Payload capacity: 5kg+ (sufficient for all electronics) - IP54 dust protection for quarry environment

3.5 GPS — RTK GPS Module (₹8,000)

- Standard GPS:** ±2-5m accuracy (insufficient for quarry block mapping)
- RTK GPS (e.g., SparkFun RTK Express):** ±2cm accuracy
- Each block measurement tagged with centimeter-accurate coordinates
- Enables quarry-wide block inventory mapping
- Prevents GPS spoofing through RTK correction signals

3.6 Lighting — LED Panel Array (₹2,000)

- Controlled, consistent illumination for photography
- Eliminates shadow artifacts that confuse depth estimation
- Essential for quarry pits with variable lighting
- Ring-mounted around camera for uniform illumination

4. DRISHTI Measurement Pipeline on Robot

4.1 Autonomous Quarry Survey Workflow

AUTONOMOUS SURVEY MODE

Step 1: NAVIGATE TO QUARRY ZONE

Robot navigates to designated quarry survey area via RTK GPS waypoints.

LiDAR provides obstacle avoidance around quarry equipment.

Step 2: DETECT GRANITE BLOCKS

OAK-D Pro scans surroundings.

YOLOv11 detects and segments individual granite blocks.

Each block assigned a unique Block-ID.

Step 3: APPROACH EACH BLOCK

Robot navigates to optimal measurement position (1-2m from block face).

Multiple viewpoints captured (front, side, top if accessible).

Step 4: MEASURE DIMENSIONS

Stereo depth → dense point cloud of block surface.

Segment block from background (ground plane removal).

Fit bounding box → extract Length, Width, Height.

Compute volume via convex hull on point cloud.

Cross-validate with ToF laser measurement.

Step 5: CLASSIFY & COMPUTE FEE

Volume → Above Gangsaw (>2.5 m³) or Below Gangsaw.

Map to granite category per AP government schedule.

Compute seigniorage fee automatically.

Step 6: LOG & MOVE TO NEXT BLOCK

Store: RGB photo + depth map + dimensions + GPS + fee + confidence score.

Move to next detected block. Repeat Steps 3-5.

Step 7: GENERATE SURVEY REPORT

After all blocks measured, compile quarry survey report.

Transmit to DRISHTI cloud platform via WiFi/4G.

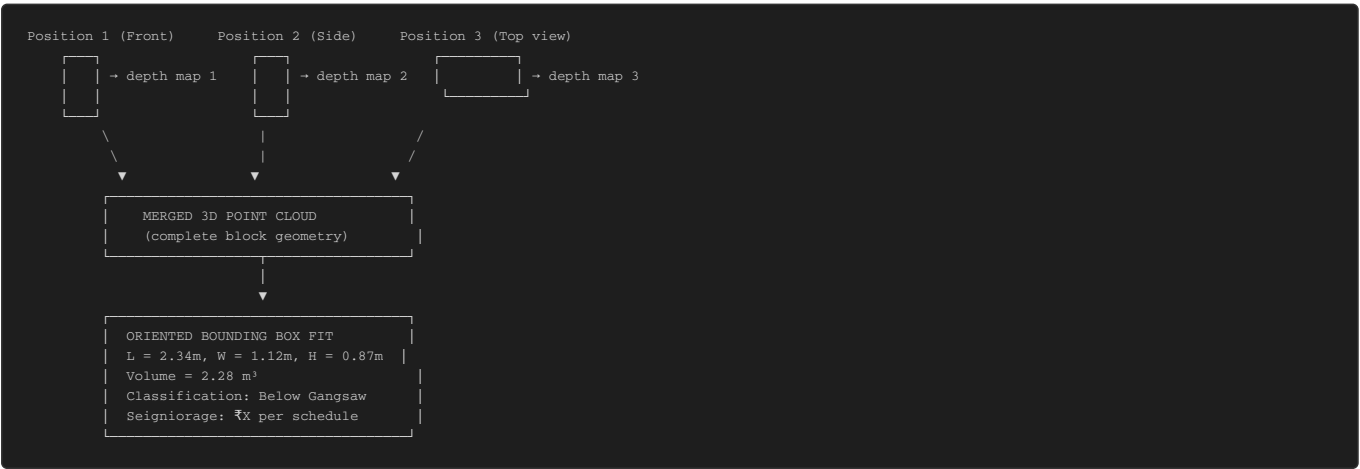
4.2 Measurement Accuracy Comparison

Method	L × W × H Accuracy	Volume Accuracy	Time per Block	Human Effort
Manual tape measure	±2cm	±5%	5-10 min	2 officers
Mobile phone (DRISHTI v1)	±10-20cm	±15-20%	1-2 min	1 officer + phone
Robot (this design)	±2-3cm	±3-5%	30-60 sec	Zero (autonomous)

5. 3D Point Cloud Measurement Approach

Unlike DRISHTI's monocular approach that estimates a single depth map from one photo, the robot captures full 3D geometry:

Multi-View Reconstruction



Advantages Over Single-Photo Approach:

1. **Complete geometry** — captures all visible faces, not just front
2. **Handles irregular shapes** — point cloud captures curves and chips
3. **No reference marker needed** — stereo baseline provides absolute scale
4. **Shadow-proof** — robot-mounted LEDs eliminate lighting issues
5. **Repeatable** — same measurement process every time (no human variability)

6. Handling DRISHTI's Known Challenges

DRISHTI Risk	How Robot Solves It
Variable quarry conditions (dust, lighting)	Robot-mounted LED panels + IR structured light on OAK-D
Irregular block shapes	Point cloud + convex hull volume (not L×W×H approximation)
GPS spoofing	RTK GPS with correction signals + robot odometry cross-check
Network connectivity at quarries	Full offline processing; stores results, syncs when connected
Calibration accuracy	Hardware stereo = self-calibrated; no external marker needed
Wet/reflective surfaces	IR dot projector provides depth even on wet granite
Human measurement variability	Eliminated — robot measures identically every time

7. Integration with DRISHTI Cloud Platform

The robot operates as a field measurement device that feeds into DRISHTI's existing architecture:



API Payload from Robot (per block):

```
{
  "block_id": "QRY-AMR-2026-0847",
  "quarry_id": "APQRY-0023",
  "timestamp": "2026-08-26T10:34:12+05:30",
  "gps": {"lat": 16.5734, "lon": 80.3567, "accuracy_cm": 2},
  "dimensions": {
    "length_m": 2.34,
    "width_m": 1.12,
    "height_m": 0.87,
    "confidence": 0.94
  },
  "volume_m3": 2.28,
  "measurement_method": "stereo_pointcloud_multiview",
  "classification": "below_gangsaw",
  "seigniorage_fee_inr": 4560,
  "evidence": {
    "rgb_images": ["front.jpg", "side.jpg", "top.jpg"],
    "depth_maps": ["front_depth.png", "side_depth.png"],
    "pointcloud": "block_0847.ply"
  },
  "robot_metadata": {
    "robot_id": "DRISHTI-BOT-01",
    "battery_percent": 72,
    "measurement_duration_sec": 45
  }
}
```

8. DRISHTI-Specific Bill of Materials

Component	Purpose (DRISHTI Context)	Cost (₹)
Jetson Orin Nano Super	YOLOv11 block detection + stereo depth processing	21,000
OAK-D Pro	Hardware stereo depth (±2cm accuracy) + 4K evidence photos	16,800
RPLiDAR A2	Autonomous navigation through quarry	16,800
Tracked chassis + motors	Quarry terrain traversal (gravel, dust, uneven)	10,000
ESP32-S3 + motor drivers	Motor control + odometry	2,500
RTK GPS module	Centimeter-accurate block location tagging	8,000
ToF laser sensor (TFMini-S)	Cross-validation of stereo depth	2,500
LED panel array	Consistent illumination for measurement	2,000
4G LTE module	Data sync to DRISHTI cloud	2,000
3S LiPo (10000mAh)	Extended quarry survey time (~3 hrs)	5,000
DC-DC converters + wiring	Power distribution	2,100
Dust/weather enclosure (IP54)	Quarry dust protection	2,500
Misc (mounts, connectors)	Assembly	2,000
TOTAL		₹93,200

9. Performance Targets (Aligned to DRISHTI Success Metrics)

DRISHTI Metric	Mobile App Target	Robot Target	Improvement
Dimension accuracy	>95% (within 5cm)	>98% (within 2-3cm)	2× better
Volume accuracy	>90%	>97% (point cloud)	Significant improvement
AI confidence score	>85% for 80% of blocks	>92% for 95% of blocks	Better consistency
Physical inspection reduction	70%	95% (fully autonomous)	Near-full automation
Processing time per block	<30 seconds (upload+cloud)	<60 seconds (multi-view, but no upload)	Comparable, richer data
OMEPS 2.0 sync	100%	100% (same API)	Same

Robot-Specific Metrics:

- Blocks measured per hour: 40-60 (autonomous, no human needed)
- Quarry coverage: Full site survey in 2-3 hours
- Terrain handling: Traverses gravel slopes up to 15°
- Dust resistance: Operates in standard quarry dust conditions (IP54)
- Repeat measurement variance: <1cm (eliminates human variability)

10. Operational Workflow for DMGO Officers

Before Robot (Current):

1. Officer drives to quarry (30-60 min)
2. Manually photographs each block with phone (2 min/block)
3. Waits for cloud processing (30 sec/block)
4. Reviews AI estimates, manually verifies suspicious ones
5. Approves measurements, fee computed
6. Total: 4-6 hours for 50-block quarry, requires 1-2 officers on-site

With Robot (Proposed):

1. Robot deployed at quarry (can be pre-positioned or driven)
2. Officer triggers survey from dashboard (remote or on-site)
3. Robot autonomously navigates, detects, and measures all blocks
4. Results stream to DMGO dashboard with evidence
5. Officer reviews high-confidence results (approve in batch)
6. Officer visits only for low-confidence measurements (<5% of blocks)
7. Total: 2-3 hours (robot autonomous), officer reviews remotely in 30 min

Efficiency gain: 80% reduction in officer field time

11. Development Roadmap

Phase	Duration	Deliverables
Phase 1: Block Detection	Week 1-2	YOLOv11 trained on granite blocks, runs on Jetson
Phase 2: Stereo Measurement	Week 3-4	OAK-D depth → point cloud → dimension extraction
Phase 3: Multi-View Fusion	Week 5	Robot captures multiple angles, fuses into single 3D model
Phase 4: Quarry Navigation	Week 6-7	Tracked chassis, SLAM on quarry terrain, waypoint navigation
Phase 5: Fee Engine + API	Week 8	Seigniorage computation, DRISHTI platform API integration
Phase 6: Field Validation	Week 9-10	Test at real quarry, compare vs manual measurements

12. Sources

- [NVIDIA Jetson Orin Nano Super Developer Kit](#)
- [Luxonis OAK-D Pro — Stereo Depth Documentation](#)
- [RealSense vs OAK-D for Robotics Depth](#)
- [Empirical Comparison of Depth Cameras for Robotics](#)
- [RPLiDAR A2 with ROS2 Cartographer](#)
- [YOLOv11 Object Detection](#)

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